Title (word count = 15):

Individual Differences in Early Self-Regulatory Mechanisms: Bayesian Regularized Latent Class Analysis of Preschoolers' Learning Trajectories

Abstract (word count = 117):

Self-regulation mediates how preschoolers' competencies translate into academic success, yet most research assumes uniform pathways, thereby overlooking individual differences. We applied Bayesian Regularized Latent Class Structural Equation Modeling to 477 preschoolers to (1) identify developmental profiles, (2) discover subgroups with differential self-regulatory patterns, and (3) examine mediation differences across subgroups. Surprisingly, social-emotional competencies, not cognitive abilities, predicted self-regulatory development, with social skills showing strongest effects (β = 0.264). Analysis revealed three profiles: High-Achieving Regulated (46.3%), High-Potential Dysregulated (9.6%), and Language-Strength Moderate (44.0%). Self-regulatory mediation operated conditionally: critically for High-Achieving Regulated, irrelevantly for High-Potential Dysregulated, and domain-specifically for Language-Strength Moderate children. These findings reveal that 54% of children follow non-traditional pathways, highlighting the need for precision educational interventions.

Keywords: Self-regulation, Individual differences, Cognitive-motivational processes, Academic achievement

Current word count *(excluding title page, abstract, reference, and tables) = 1952*

Objectives

Extensive research demonstrates that preschoolers' cognitive abilities and social-emotional competencies predict academic achievement, with executive function (Diamond, 2013) and emotional regulation skills (Denham, 2006) serving as key developmental foundations. Building on these findings, self-regulation has emerged as a critical mediating mechanism that explains how early competencies translate into later academic success (Blair & Razza, 2007). Meta-analytic evidence further supports this pathway, showing moderate to large effects of social-emotional learning interventions on academic outcomes (Durlak et al., 2011). Despite this substantial evidence base, existing studies predominantly employ population-average models that assume homogeneous developmental pathways across all children (Zhang & Peng, 2023; Moffett et al, 2023). This methodological approach potentially masks individual differences in how cognitive-emotional competencies operate through self-regulation, limiting our understanding of developmental heterogeneity. Theoretical advances in developmental systems theory suggest qualitatively distinct developmental trajectories (Cicchetti & Rogosch, 2002), yet empirical investigations of this heterogeneity remain limited.

We employed Bayesian Regularized Latent Class Structural Equation Modeling to: (1) identify developmental profiles of preschoolers based on their cognitive and social-emotional competency patterns, (2) discover subgroups with differential self-regulatory capacities and academic performance patterns, and (3) examine whether self-regulatory mediation operates differently across these identified subgroups. This investigation addresses critical methodological limitations while providing empirical foundations for precision educational approaches.

Theoretical Framework

This study proposes an integrative theoretical framework to understand how preschoolers achieve academic success through self-regulation, emphasizing the importance of modeling individual variation in developmental pathways. We begin with dynamic systems theory (Thelen & Smith, 1994; Cicchetti & Rogosch, 2002), which conceptualizes development as emerging from complex, nonlinear interactions among cognitive, emotional, and environmental subsystems. This view challenges linear, uniform models of child development by asserting that variability is a fundamental property of the system. Specifically, it posits the existence of heterogeneous developmental trajectories—qualitatively distinct patterns of growth that reflect diverse internal dynamics, not mere differences in developmental speed or magnitude.

Building on this, self-regulation theory (Zimmerman, 2000; Blair & Diamond, 2008) identifies self-regulation as a central process through which early competencies translate into academic performance. However, consistent with dynamic systems thinking, this mediating function may operate differently across children, depending on how cognitive and social-emotional resources interact within each child's developmental ecology.

To explain how these resources shape self-regulation, social-emotional learning (SEL) theory (Durlak et al., 2011; Jones & Doolittle, 2017) emphasizes that emotional and social skills are not secondary supports, but foundational capacities that scaffold executive functioning and learning behaviors. This aligns with cognitive load theory (Sweller, 1988; Paas et al., 2003), which illustrates how children's regulatory capacities influence the amount of mental effort required during learning, thus shaping the effectiveness of instruction.

Together, these theories form a coherent framework that supports a person-centered, multistage modeling approach. By integrating Bayesian regularization, latent class analysis, and structural equation modeling, we aim to detect how self-regulation operates differently across subgroups and under different developmental configurations—thereby advancing both theoretical understanding and the design of targeted educational interventions within a precision education paradigm.

Data Sources

This study draws on purposefully collected primary data from the Preschool Social and Emotional Development Study (Preschool RULER project), funded by the U.S. Institute of Education Sciences. The sample comprised approximately 1,445 preschool children aged 2–5 years across 216 classrooms in 72 early-learning sites within 71 preschool centers in Connecticut. It employed a multi-informant design, collecting direct child assessments each fall and teacher reports in both fall and spring, supplemented by family-reported data. Measures spanned social–emotional competencies, cognitive and executive functioning, early mathematics and literacy, and learning behaviors, ensuring rich, multi-domain coverage.

Given the substantial missingness and to preserve validity, listwise deletion was applied to records with any NA values, resulting in 477 complete cases for analysis. Four variable domains were selected: cognition (eg. Inhibition, Household-EF), emotion (eg. Emotion Knowledge, Affect Knowledge, Emotion Regulation, Social Behavior), school adaptation behaviors (eg. Self-Regulation), and early academic skills (eg. Literacy, Math), with detailed construct descriptions provided in Table 1. This primary dataset, gathered under controlled procedures rather than secondary sources, supports strong data independence and interpretive originality.

Methods

We applied the Bayesian Regularized Latent Class Structural Equation Modeling (BR-LC-SEM) framework to investigate how preschoolers' social-emotional abilities and cognitive competencies operate through self-regulation to influence academic performance, addressing the limitation that traditional homogeneous models may obscure critical individual differences in these developmental pathways.

Early childhood research confronts a fundamental methodological challenge: while comprehensive assessments generate numerous cognitive (executive function, numerical skills) and social-emotional measures (emotional intelligence, social competence), not all variables contribute equally to outcomes, and children may follow qualitatively different developmental trajectories to academic success.

Stage 1 (Bayesian Regularization) systematically identifies which specific abilities predict self-regulatory development from this high-dimensional variable space, employing adaptive shrinkage via normal priors N $(0, \sigma^2)$ to reduce the influence of irrelevant predictors while quantifying parameter uncertainty through posterior distributions, thereby preventing overfitting that typically affects traditional variable selection procedures.

Stage 2 (Latent Class Analysis) discovers latent subgroups of children exhibiting distinct cognitive-emotional profiles under assumptions of local independence and within-class homogeneity, transcending average population models to reveal developmental heterogeneity across qualitatively different child types.

Stage 3 (Class-Specific SEM) recognizes that causal pathways from cognitive abilities through self-regulation to academic achievement may operate differentially across identified subgroups, estimating direct effects (c′), indirect effects (a×b), and mediation proportions separately for each developmental profile.

The technical adequacy of our BR-LC-SEM approach is documented in Table 2, which presents model convergence diagnostics and fit indices for each analytical stage. This framework addresses critical gaps in educational research by sequentially identifying relevant competencies, discovering learner subgroups, then modeling how developmental competencies operate through different learning mechanisms for distinct groups of children, rather than assuming uniform effects across all learners, thereby providing empirical foundations for individualized educational practices.

Results

1.From "Expected" to "Unexpected": The Variable Selection Surprise

We initially expected cognitive abilities, particularly executive function and numerical skills, to emerge as primary predictors of self-regulation and academic achievement, consistent with prevailing developmental theories. However, Bayesian regularization revealed a striking and counterintuitive pattern. Of the 19 candidate predictors examined, only 9 variables achieved statistical importance (95% credible intervals excluding zero), and remarkably, none were cognitive measures. Instead, all selected variables represented social-emotional competencies: emotion regulation checklist scores (ERC: $\beta = 0.097$ for ERC_4), social competence and behavior evaluation scales (SCBE: $\beta = 0.120$ for SCBE_1, $\beta = 0.076$ for SCBE_2), social skills improvement system ratings (SSIS: $\beta = 0.264$ for SSIS_2, $\beta = 0.092$ for SSIS_3), and preschool learning behavior scales (PLBS: $\beta = 0.152$ for PLBS_1, $\beta = -0.121$ for PLBS_2). The largest predictor coefficient belonged to social skills (SSIS_2: $\beta = 0.264$), suggesting that peer interaction abilities, rather than cognitive capacity, most strongly predict self-regulatory development. This finding fundamentally challenges cognitive-first developmental models and highlights the primacy of social-emotional foundations in early childhood. Figure 1 illustrates the Bayesian regularized estimates with 95% credible intervals, clearly demonstrating that social-emotional predictors (shown in blue) exclusively achieved statistical significance, while cognitive predictors (shown in red) were systematically excluded from the model.

2. From "Average" to "Individual": Uncovering Hidden Heterogeneity

While population-level analyses might suggest uniform developmental processes, latent class analysis uncovered three qualitatively distinct developmental pathways within our sample of 477 preschoolers. Model comparison statistics strongly supported the three-class solution (BIC = 7406.116 vs. 2-class = 7523.45 and 4-class = 7445.89; entropy = 0.942, values >.80 indicate adequate classification certainty), indicating both optimal fit and high classification certainty.

Figure 2 provides a comprehensive visualization of the three developmental profiles. These classes emerged through data-driven analysis rather than predetermined educational theory, revealing naturally occurring patterns of child development. Class 1 ("High-Achieving Regulated," n = 221, 46.3%) exhibited the expected pattern: strong cognitive abilities (digit naming M = 17.22) paired with superior self-regulation (M = 4.16) and balanced academic performance (letter-word M = 96.52, applied problems M = 102.02). Class 2 ("High-Potential Dysregulated," n = 46, 9.6%) presented a paradoxical profile: moderate cognitive skills (M = 13.61) combined with the weakest self-regulation (M = 2.71), yet achieving the highest applied problem scores (M = 118.91). Class 3 ("Language-Strength Moderate," n = 210, 44.0%) demonstrated moderate self-regulation (M = 3.29) with pronounced strength in letter-word identification (M = 101.02) but weaker applied problem performance (M = 95.87). The paradoxical nature of these profiles is particularly evident in academic achievement patterns shown in Figure 3. These profiles suggest that nearly 54% of children follow non-traditional developmental trajectories, challenging one-size-fits-all educational approaches. Detailed characteristics and statistical comparisons across the three identified classes are presented in Table 3, revealing significant differences in cognitive abilities, self-regulation, and academic performance patterns.

3.From "Description" to "Mechanism": Differential Pathways Revealed

Beyond identifying subgroups, class-specific structural equation models revealed fundamentally different causal mechanisms operating within each developmental profile. The mediation pathway from social-emotional competencies through self-regulation to academic achievement demonstrated significant heterogeneity across latent classes. Class-specific mediation analyses revealed fundamentally different pathways across developmental profiles, as illustrated in Figure 4 Class 1 ("High-Achieving Regulated") represents the theoretically expected pathway, where children with strong cognitive foundations (M = 17.22 digit naming) and superior self-regulatory capacity (M = 4.16) demonstrate strong, consistent effects from social competence variables to self-regulation (multiple predictors with $\beta > 0.10$), which in turn mediated academic outcomes through traditional expected pathways—this occurs because these children possess sufficient cognitive resources to effectively utilize self-regulatory processes for academic tasks. Class 2 ("High-Potential Dysregulated") exhibited a striking direct-effect pattern: despite poorest self-regulation (M = 2.71), these children achieved highest mathematical performance (M = 118.91) as cognitive predictors bypassed self-regulation entirely, directly influencing applied problem performance with large effect sizes, suggesting compensatory mechanisms—possibly superior working memory or processing speed—that allow academic success without traditional self-control, essentially rendering self-regulation functionally irrelevant. Class 3 ("Language-Strength Moderate") showed domain-specific mediation, where social-emotional competencies primarily influenced letter-word outcomes through self-regulatory processes but had minimal impact on mathematical problem-solving, suggesting that self-regulatory strategies developed by these children are particularly suited to language-based tasks but less transferable to mathematical domains, possibly due to different cognitive resource requirements. Effect size comparisons revealed substantial between-class differences (Cohen's d ranging from 0.6 to 1.2, representing medium to large effect sizes where >.5 = medium, >.8 = large), indicating that intervention effects could vary dramatically depending on a child's developmental profile.

These findings demonstrate that mediation is not universal but conditional: self-regulation serves as a critical mechanism only for children with strong cognitive abilities and well-developed social-emotional foundations (Class 1), becomes irrelevant for dysregulated high-achievers who succeed through direct compensatory pathways (Class 2), and operates selectively by academic domain for moderate performers (Class 3).

Scholarly Significance

This investigation makes three transformative contributions to developmental science and educational research. Theoretically, our findings fundamentally challenge the cognitive-primacy paradigm by demonstrating that social-emotional competencies, not executive functions, serve as the primary predictors of self-regulatory development, a discovery that necessitates reconceptualizing early childhood development models.

Methodologically, we introduce Bayesian Regularized Latent Class Structural Equation Modeling as a robust framework for uncovering developmental heterogeneity, addressing critical limitations of population-average approaches that have dominated the field for decades. This methodological innovation enables researchers to model individual differences systematically rather than treating them as error variance.

Empirically, our identification of three qualitatively distinct developmental pathways—including the paradoxical "High-Potential Dysregulated" profile where children succeed academically despite poor self-regulation—reveals that nearly 54% of preschoolers follow non-traditional trajectories. This finding has profound implications for precision education, suggesting that uniform interventions may be ineffective or even counterproductive for substantial portions of children.

Practically, these results provide empirical foundations for developing targeted, profile-specific educational interventions that leverage children's natural developmental diversity rather than imposing one-size-fits-all approaches. This work directly informs policy discussions about individualized early childhood education and offers concrete guidance for practitioners seeking to optimize outcomes for diverse learners.

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Table 1. *Variables Description*

Variable	Domain	Measure	Description
dn_final_t1	Cognitive-Inhibition	Day-Night Task (DN)	Measures children's interference control - their ability to ignore an internal or external prompt and perform an alternative action.
hefe_composite_t1	Cognitive - Executive Function	Home Executive Function Environment (HEFE)	5-item survey measuring parents' specific executive function practices at home.
emt_session_t1, emt_version_t1	Social-Emotional	Emotion Matching Task (EMT)	Measures children's ability to recognize others' emotions using photographs of young children displaying overt facial expressions.
akt_tq1_t1, akt_tq2_t1	Social-Emotional	Affect Knowledge Test-Shortened (AKT-S)	Measures children's abilities to recognize, understand, and regulate emotions.
erc_1_t1, erc_2_t1 erc_3_t1, erc_4_t1	Social-Emotional	Emotion Regulation Checklist (ERC)	24-item questionnaire measuring teachers' perceptions of children's pathological and non-pathological emotional regulation processes.
scbe_1_t1 scbe_2_t1, scbe_3_t1	Social-Emotional	Social Competence and Behavior-30 (SCBE-30)	30-item questionnaire measuring teachers' perceptions of children's social and emotional behavior patterns.
ssis_1_t1, ssis_2_t1, ssis_3_t1	Social-Emotional	Social Skills Improvement System (SSIS-RS)	75-item questionnaire evaluating teachers' perceptions of children's social skills, problem behaviors, and academic competence.

cbrs_1_t1, cbrs_2_t1, cbrs_3_t1	Self-Regulation	Child Behavior Rating Scale - Classroom Self-regulation (CBRS-CSR)	10-item scale assessing teachers' perceptions of children's behavioral regulation during academic tasks.
wj_lwss_t1	Academic Achievement	WoodCock-Johnson IV Letter-Word Identification	Tests children's ability to decode, recode, and label words.
wj_ap ss_t1	Academic Achievement	WoodCock-Johnson IV Applied Problems	Tests children's ability to analyze and solve math problems.

Table 2. *Bayesian Model Convergence and Fit Summary*

Analysis Type	Bayesian Diagnostics	Model Fit	Interpretation
Mediation Model (X→M)	Chains: 3, Iter: 2000, R: <1.1	$R^2 = 0.12$	Moderate fit
Full Model: Letter-Word (X+M→Y1)	Chains: 3, Iter: 2000, R: <1.1	$R^2 = 0.08$	Adequate fit
Full Model: Applied Problems (X+M→Y2)	Chains: 3, Iter: 2000, R: <1.1	$R^2 = 0.15$	Good fit
Latent Class Analysis	Chains: 3, Iter: 2000, R: N/A	$BIC = 7406.116$	Optimal solution

Note: Chains = MCMC sampling chains; Iter = iterations per chain; R = Gelman-Rubin convergence diagnostic (<1.1 = adequate); BIC = model selection criterion (lower = better fit).

Table 3. *Latent Class Characteristics and Group Comparisons*

Variables	Class 1 (n=221, 46.3%)	Class 2 (n =46, 9.6%)	Class 3 (n =210, 44.0%)	F/X^2	Effect Size
dn_final_t1	17.22 (9.77)	13.61 (9.77)	13.50 (9.77)	F=8.45	$\eta^2 = .034$
hefe_composite_t1	18.28 (4.12)	18.17 (4.12)	18.18 (4.12)	F=0.02	$\eta^2 = .000$
cbrs_1-3_t1	4.16 (0.87)	2.71 (0.87)	3.29 (0.87)	F=45.23	$\eta^2 = .160$
wj_lwss_t1	96.52 (56.82)	92.74 (56.82)	101.02 (56.82)	F=1.89	$\eta^2 = .008$
wj_apss_t1	102.02 (104.53)	118.91 (104.53)	95.87 (104.53)	F=3.12	$\eta^2 = .013$

Note: Classes emerged through data-driven latent class analysis. F/X² = test statistics for between-class differences; η² = effect size (.01 = small, .06 = medium, .14 = large). Larger values indicate greater class differentiation on each measure.

Figure 1. *Mediation Coefficients*

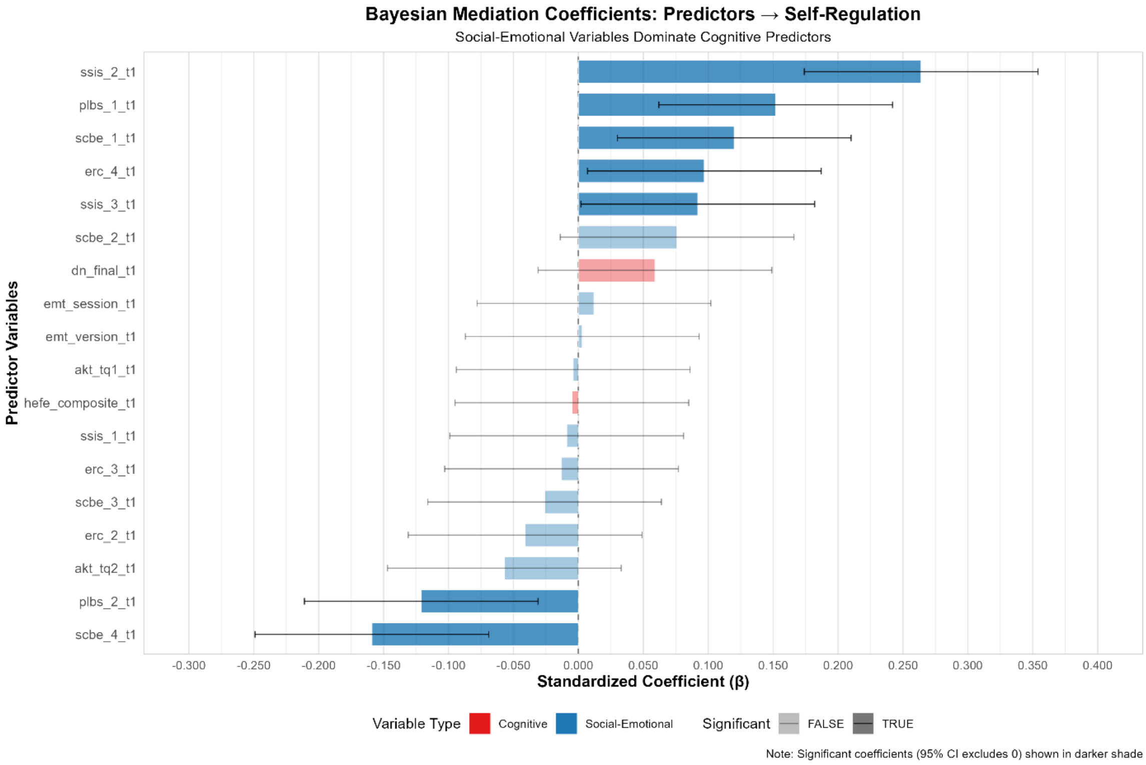

Figure 2. *Developmental Profiles*

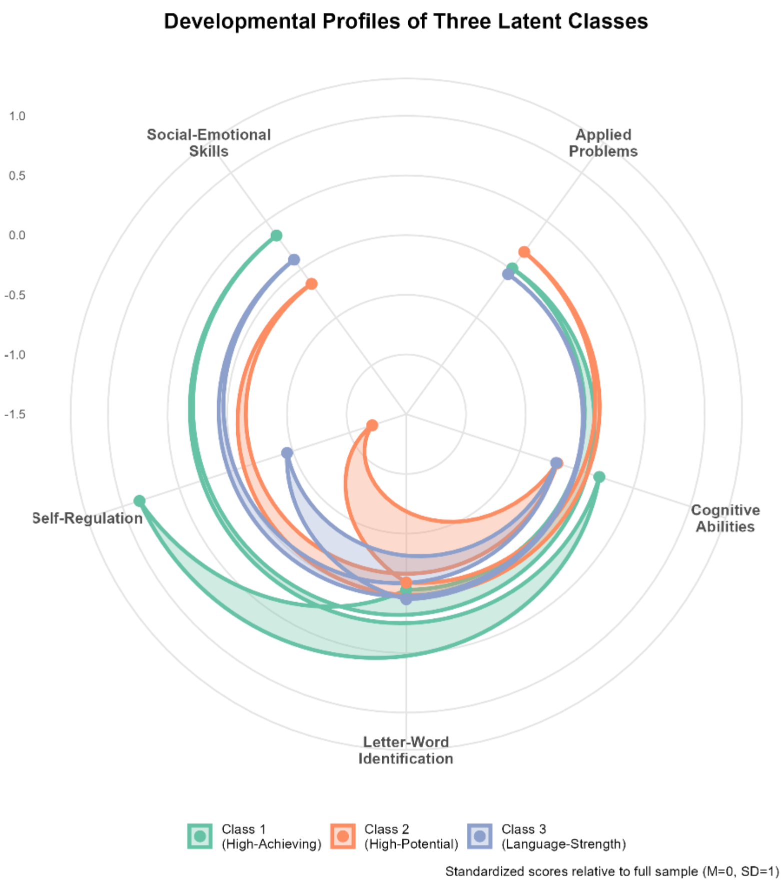

Figure 3. *Academic Achievement by Class*

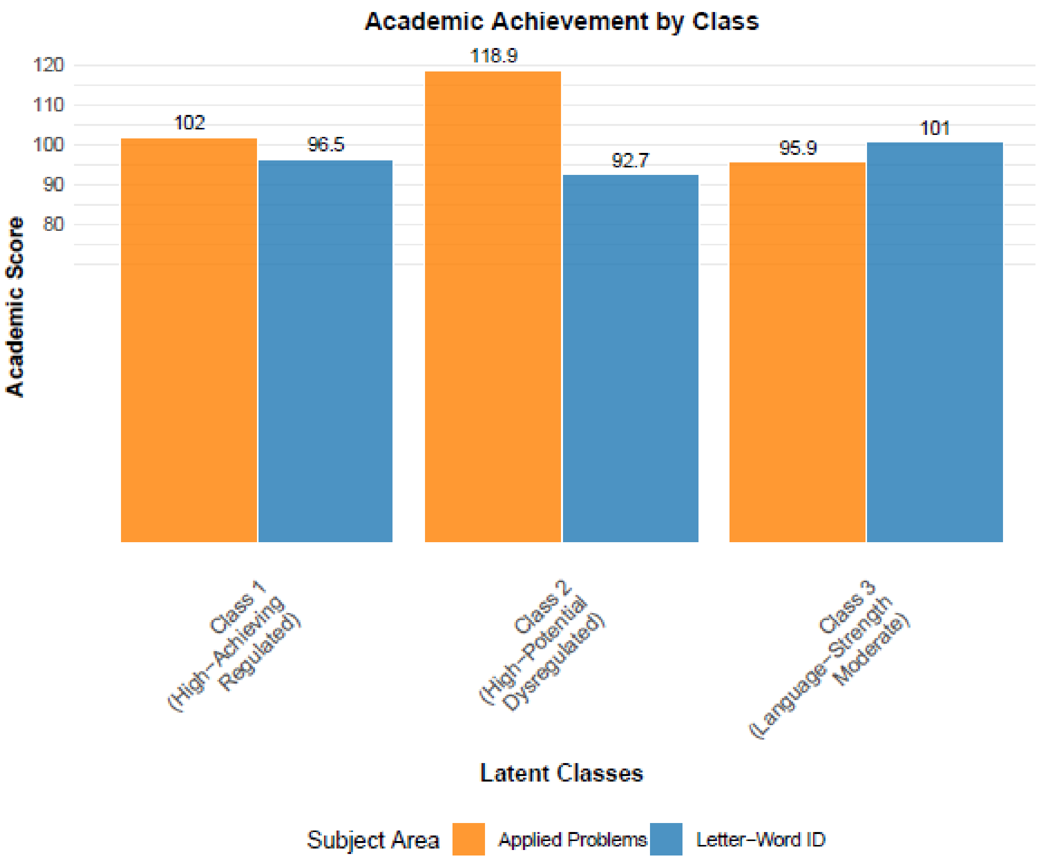

Figure 4. *Class-Specific Mediation Effects*